



Okay, so I have a literature review task on my university thesis. So while selecting the topic, we have stumbled upon the given things below. So now I don't know where to start and what to do and which papers to look at and what should be our target but what I know is I want to make a world-class paper that actually solves a problem either in depth or in breadth. So I want you to do a long-running research to find me the best papers That I can take reference from and build something on top of them.

Title:

Domain-Generalizable WiFi CSI Based Surveillance and Vital Sign Monitoring: A Framework for Movement Tracking, Security Detection, and Biometric Identification

Abstract:

This research will prepare a custom dataset by emulating real-world conditions across various activities, forming the basis of a surveillance framework built on continuous WiFi Channel State Information (CSI) acquisition.

The thesis will investigate the monitoring of residential environments for security and welfare purposes. Movement patterns of occupants, particularly elderly individuals, will be tracked to assess well-being, with prolonged absence of movement triggers. Health status indicators will also be derived continuously from the collected data.

This research will further explore the framework's use as a security mechanism, detecting unauthorized presence, while estimating occupant location and count.

Additionally, this thesis will try to develop a biometric identification module using heartbeat and breathing-pattern signatures from CSI data to establish unique physiological profiles, enabling identification of intruders.

Finally, this research will evaluate the feasibility of deploying the proposed framework on low-power, computationally constrained edge devices.

Your current idea contains several thesis-sized problems. The strongest, most defensible contribution is to make **domain-generalizable, uncertainty-aware WiFi CSI monitoring for home safety**, rather than claiming a single system can simultaneously deliver clinical vital-sign monitoring, reliable intruder biometrics, tracking, counting, and edge deployment.

The literature gap is real: CSI systems often fail across new rooms, device configurations, people, orientation, and furniture layouts. A strong thesis can solve this through a rigorous cross-domain dataset and evaluation protocol, then show which safety tasks remain reliable under deployment shift. ^[1] ^[2]

Recommended thesis focus

Better title

Domain-Generalizable WiFi CSI Sensing for Home Safety: Cross-Room Presence, Motion, Respiration, and Open-Set Occupant Recognition on Edge Devices

This wording is stronger because it:

- Makes generalization—the true hard problem—the central scientific contribution.
- Separates **safety sensing** from unsupported claims of clinical diagnosis.
- Uses “open-set occupant recognition,” which is the correct security formulation: identify a known enrolled occupant *or reject an unknown person*, rather than simply forcing every person into a known identity class.
- Avoids claiming heartbeat-based biometrics before establishing that a weak cardiac CSI component is recoverable robustly across rooms and days.

Core research question

Can one lightweight CSI representation, trained across diverse homes and device placements, reliably estimate home-safety states in a previously unseen room—and abstain when CSI quality is inadequate?

Your main outputs should be:

1. Occupancy and movement

- Empty / occupied.
- Motion / no motion.
- Long inactivity alert only after occupancy has been established and context rules are met.
- This is practical and high value. WiDetect is a major baseline: it links CSI autocorrelation to movement and aims for calibration-free, commodity-WiFi motion detection. ^[3] ^[4]

2. Respiration-aware stationary presence

- Detect whether a stationary occupied state contains a usable respiration signal.

- Estimate respiratory rate only when a **signal-quality indicator** says the estimate is trustworthy.
- Treat heart-rate estimation as an optional secondary experiment, not a core deliverable. Heartbeat is much weaker than respiratory motion and is easily masked by respiration and environmental noise. ^[5] ^[6]

3. Room-level zone or trajectory

- Start with zone-level localization: e.g., entrance / bed zone / desk zone.
- Pursue continuous trajectory reconstruction only if you can install multiple receiver viewpoints. WiTraj's reported sub-meter tracking depends on three WiFi links and explicitly fuses views to handle orientation and walking direction. ^[7]

4. Open-set authorization

- Known resident versus unknown person, only during a walking or designated enrollment sequence.
- Do not make “heartbeat and breathing-pattern signatures identify intruders” your main claim. Existing CSI identity work is mainly based on gait, movement signatures, or respiration patterns in controlled settings; it is not proof that passive heartbeat is a stable household biometric. ^[8] ^[9] ^[10]

5. Edge feasibility

- Report latency, memory, energy/power, model size, throughput, and privacy/data-retention design.
- ESP32 hardware exposes CSI and supports human-presence/motion examples, so it is viable for collection and coarse sensing; a Raspberry Pi is the safer first platform for signal processing and ML inference. ^[11]

What is realistically achievable

Proposed capability	Feasibility	Recommended scope	Thesis claim to avoid
Motion and intrusion detection	High	Primary task: motion/presence with false-alarm control	“Detects all intruders through walls”
Empty vs occupied room	High–medium	Primary task, including quiet sitting and different days	“Works universally without calibration”
Prolonged no-movement welfare alert	Medium	Rule-based alert after validated presence, with alert-delay analysis	“Detects medical emergencies”
Zone-level localization	Medium	3–5 semantic zones or room-level location	Exact continuous tracking with one link
People count	Medium–low	Optional: 0 / 1 / 2+ rather than exact counts	Accurate arbitrary crowd count in homes
Respiration rate	Medium	Stationary single-person, quality-gated, reference sensor validated	Medical-grade monitoring from all positions
Heart rate	Low–medium	Exploratory, stationary single person, clear limitations	Reliable household biometric or diagnosis

Proposed capability	Feasibility	Recommended scope	Thesis claim to avoid
Biometric identity	Medium for gait	Open-set gait/behavior identity during walking	Identity from passive heartbeat/breathing alone
Multi-person vital signs	Low	Future work or narrowly controlled extension	Accurate per-person vital signs in free-living homes
Edge deployment	High	Quantized small model; compare ESP32 and Raspberry Pi	“Low power” without measured latency/energy

CSI-Bench is especially relevant because it already frames WiFi sensing as a multitask, real-world problem: it contains more than 460 hours from 35 users, 26 environments, and 16 device types, with tasks including breathing monitoring, fall detection, localization, motion-source recognition, identity, activity, and proximity. Use it as an external benchmark and motivation—not as a replacement for your own domain-diverse home-safety dataset. [\[12\]](#) [\[13\]](#)

Best papers to read

I created a CSV literature matrix with 23 papers/datasets, direct sources, priority, contribution, and limitations to check. Start with the following 12 in this order.

Order	Paper	Why it matters to your thesis
1	<i>A Survey on Wi-Fi Sensing Generalizability</i>	Your conceptual map for domain shifts: device, person, room, placement, orientation, data augmentation, domain alignment, few-shot and continual adaptation. [1]
2	<i>A Survey on Vital Signs Monitoring Based on Wi-Fi CSI Data</i>	Establishes the signal-processing vocabulary and explains why results commonly degrade with multiple people, distance, position, and environment. [14]
3	<i>Widar3.0: Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi</i>	Seminal physics-aware cross-domain reference. It uses a body-coordinate velocity profile rather than raw CSI and evaluates across users, locations, orientations, and rooms. [15]
4	<i>AirFi: ...Unseen Environment via Domain Generalization</i>	Direct domain-generalization baseline: learns from multiple source environments without requiring target-room data during training. [16]
5	<i>WiDetect: Robust Motion Detection with a Statistical Electromagnetic Model</i>	Your primary motion/security baseline; simple, interpretable, and closer to a deployable safety detector than activity classifiers. [3]
6	<i>FarSense: Pushing the Range Limit of WiFi-based Respiration Sensing</i>	Important CSI-ratio method using two antennas; directly relevant for robust respiration sensing and phase/amplitude complementarity. [17]
7	<i>PhaseBeat: On CSI-Based Vital Sign Monitoring Using Commodity WiFi</i>	Essential classical vital-sign pipeline: inter-antenna phase difference, subcarrier selection, wavelets, stationary-state gating, breathing and heart-rate estimation. [6]
8	<i>VitalCSI: Contactless Respiratory Rate Estimation Using Consumer-Grade Wi-Fi CSI</i>	Modern hardware and stronger clinical-style protocol: 15 participants, nasal-airflow reference, 6–33 breaths/min, 30-second windows, MAE 1.20 breaths/min. [18]

Order	Paper	Why it matters to your thesis
9	PulseFi	Strong low-cost baseline for amplitude-only LSTM vital-sign estimation. Read critically: its reported split shuffles data and uses overlapping windows, so reproduce it under strict subject/session/room separation before accepting headline numbers. [5]
10	TensorBeat and MultiSense	Your multi-person respiration literature. Both reveal the central challenge: separating mixtures is not the same as assigning each signal to a specific person or identity. [19] [20]
11	CAUTION: <i>Few-Shot Open-Set Recognition</i>	Use this for the security biometric component: open-set rejection, few-shot enrollment, and gait-based authentication are much better aligned with real intruder detection than closed-set classification. [10] [21]
12	EfficientFi	Your edge/networking reference. It jointly compresses CSI at the edge and supports sensing, reporting compression from 1.368 Mb/s to 0.768 kb/s in its setup. [22]

Secondary but valuable papers

- **WiTraj** for multi-link trajectory reconstruction and the critical distinction between walking and in-place activity. [\[7\]](#)
- **BreathID** for direct prior art on respiration-based CSI identity, so your novelty cannot merely be “using breathing for identification.” [\[9\]](#) [\[23\]](#)
- **WhoFi** for a metric-learning CSI identity baseline using LSTM, BiLSTM, and Transformer encoders. Its 14-subject walking evaluation is useful, but it does not prove cross-home biometric robustness. [\[24\]](#)
- **CSI-Bench** for current in-the-wild benchmark practices. [\[12\]](#)
- **eHealth CSI** for activity, empty-room, respiration-intermittency, heart-rate, and Polar H10-related reference data. Dataset access requires a data-use agreement. [\[25\]](#)
- **Exposing the CSI** for Wi-Fi 6, bandwidth, antenna, and reproducibility limitations. [\[26\]](#)
- **Mitigating Data Leakage in a WiFi CSI Benchmark** for your evaluation section; it demonstrates that improper splits can drastically inflate reported accuracy. [\[27\]](#)
- **AM-FM** for the emerging foundation-model direction. Treat it as a research baseline or inspiration, not a replacement for your own controlled evaluation; it is a recent preprint. [\[28\]](#)

Your original novelty gap

A paper will not be world-class because it combines many modules. It needs one sharp claim that remains true under difficult conditions.

Recommended central contribution

QUALITY-DG: Quality-Gated Domain-Generalizable WiFi CSI Sensing for Home Safety

Build a single shared encoder with lightweight task heads:

CSI window $\rightarrow$ quality estimator $\rightarrow$ domain-invariant encoder $\rightarrow$ {presence, motion, zone, resp

The decisive addition is the **quality/abstention head**:

- It predicts whether a CSI window is trustworthy for a given task.
- It must be trained using observable quality markers: packet loss, RSSI/SNR, signal periodicity, motion level, subcarrier consistency, and disagreement between estimators.
- When quality is low, it produces “**unknown / insufficient evidence**” rather than a false respiration rate, a false identity, or an unsafe “all clear.”
- Report **coverage versus risk**: “when the system answers on 70% of windows, its error is X; it abstains on the remainder.” This is vastly more meaningful than forcing predictions on every window.

This is scientifically stronger than raw-domain alignment alone. AirFi uses multi-source-domain alignment and augmentation for unseen environments, while recent foundation-model work suggests broad unlabeled CSI pretraining can improve transfer; your thesis can test whether explicit sensing quality and safety-oriented abstention provide additional value on low-power hardware. ^{[16] [28]}

Publishable hypotheses

- **H1:** Multi-room domain-generalized learning improves leave-one-home-out performance versus raw CSI CNN/LSTM/Transformer baselines and versus no-domain-generalization training.
- **H2:** Quality-gated prediction lowers false alarms and high-confidence errors under environmental change more effectively than a fixed classifier threshold.
- **H3:** Self-supervised pretraining on continuous unlabeled CSI reduces the labels needed for a new home compared with training from scratch.
- **H4:** A compact, quantized model can preserve acceptable safety-task performance while meeting a defined edge latency and memory budget.
- **H5:** Gait-based open-set recognition works only in its declared operational envelope—known walkway, one person, sufficient motion—and should reject stationary or low-quality CSI windows.

Do not frame the work as clinical monitoring unless your study has appropriate ethics approval, medical-grade references, and a clinical validation protocol. VitalCSI itself emphasizes that many earlier studies had very small groups, limited breathing ranges, weak reference measures, or controlled setups. ^[18]

Dataset and protocol

Your dataset—not merely model architecture—can become your strongest contribution.

Data collection design

Collect continuous sessions, not short isolated gestures. For every session, save raw complex CSI when hardware permits, packet timestamps, RSSI, channel, bandwidth, antenna configuration, device IDs, room metadata, and synchronized ground truth.

Variation	Minimum target	Why it matters
Homes/rooms	5–8 distinct rooms, ideally 3+ homes	Main unseen-domain evaluation
Participants	20–30, diverse body sizes and ages	Cross-person generalization
Days per room/person	3–5	Captures temporal drift
Furniture states	2–3 layouts	Tests multipath sensitivity
Receiver/transmitter layouts	3–4 per room	Prevents fixed-placement overfitting
Orientations/positions	Facing, side, back; near/far	Critical for respiration and motion
Network conditions	Quiet plus ordinary WiFi interference	Deployment realism
Occupancy	0, 1, 2+	Prevents “one person only” assumptions
Activities	Still seated, still lying, walk, sit/stand, enter/exit, simulated fall only if ethically safe	Home-safety semantics
Reference signals	Polar H10/ECG for HR; respiration belt or airflow sensor for RR; manual/video event labels	Valid labels and timestamp alignment

The eHealth CSI dataset is useful as a template because it includes absence, normal and intermittent breathing, posture changes, walking, running, sweeping, and fall-related sessions. Its DS2 protocol also records Polar H10 data, but your paper must disclose that the dataset access is controlled and that smartwatch/belt references differ in precision from ECG or airflow. ^[25]

Essential labels

- occupancy: empty / occupied / uncertain
- motion: none / minor body movement / walking / non-human disturbance
- zone: entry / bed / desk / other
- count: 0 / 1 / 2+
- activity: stationary, sit-to-stand, walking, lying, etc.
- respiration waveform/rate: only when a valid reference sensor is attached
- identity: pseudonymized enrolled participant ID

- authorization: enrolled / unknown person
- CSI quality: packet loss, interference, signal-level statistics, and an annotation for major confounders
- event: entry, exit, fall simulation, long-stillness episode

Strict evaluation splits

Never use random windows from the same recording in both training and test data. Adjacent or overlapping CSI windows are highly correlated, so this leaks information.

Use these five protocols:

Protocol	Test set	What it proves
Within-session	Held-out non-overlapping segment	Only pipeline sanity check
Leave-one-session-out	Different recording session/day	Temporal robustness
Leave-one-subject-out	People absent from training	Activity/vital generalization
Leave-one-room/home-out	Entire unseen room/home	Your main domain-generalization result
Leave-device-layout-out	Unseen placement/device configuration	Deployment robustness

For identity, use a different protocol:

- **Enrollment sessions:** Day 1, known residents only.
- **Known-user test:** Different day, different room configuration.
- **Unknown-user test:** people never enrolled.
- Measure TAR, FAR, FRR, AUROC, equal-error rate, and unknown-detection AUROC—not just closed-set accuracy.

Improper participant-level partitioning has been shown to create CSI benchmark leakage and sharply inflate results because models learn individual signatures rather than general activity patterns. ^[27]

Model and system plan

Phase 1: build robust baselines

Implement these before proposing a novel model:

1. Signal-processing baseline

- Amplitude extraction and/or calibrated phase difference.
- Hampel/outlier filtering, detrending, band-pass filtering.
- Per-window subcarrier selection using SNR or periodicity.
- FFT / autocorrelation / peak-detection respiration estimator.
- PhaseBeat and FarSense are the best references here. ^{[6] [17]}

2. Task-specific ML baseline

- Random forest on hand-engineered features.
- 1D CNN.
- GRU/LSTM.
- Small Transformer.
- SenseFi provides starter implementations and comparative model choices across public datasets. ^[29]

3. Generalization baseline

- ERM: normal supervised training.
- Data augmentation only.
- MMD source-domain alignment.
- Domain-adversarial classifier.
- AirFi-style domain-generalized encoder.
- Few-shot new-room calibration as a separate setting—not mixed with zero-shot claims. ^{[1] [16]}

4. Edge baseline

- Float32, then int8 quantized versions.
- Measure parameters, RAM, flash/storage, mean/p95 latency, CSI capture rate, and energy per inference.
- Separate capture cost from inference cost.

Phase 2: your novel model

Use a multi-task architecture only where labels and mechanisms overlap:

- **Shared encoder:** amplitude or CSI-ratio spectrogram plus metadata such as packet timing and RSSI.
- **Quality head:** produces per-task confidence/abstention.
- **Safety heads:** occupancy, motion, zone, count class.
- **Physiology head:** respiration rate with uncertainty.
- **Identity head:** only on gait/motion windows; metric embedding plus open-set threshold.
- **Domain head:** adversarial or MMD-based alignment across room, layout, participant, device, and day labels.

Avoid forcing all outputs from every window:

- Walking is informative for gait identity and tracking but corrupts subtle vital signs.
- Stillness is suitable for respiration but is poor evidence for gait identity.
- Multi-person CSI may be useful for count/activity but is not automatically separable into person-specific vital signs.

A state machine will make the system scientifically cleaner:

empty $\rightarrow$ motion/entry $\rightarrow$ occupied $\rightarrow$ $\begin{cases} \text{walking: location + optional identity} \\ \text{stationary: respiration quality check} \\ \text{poor quality: abstain / request more evidence} \end{cases}$

Metrics that reviewers will respect

Safety and surveillance

- Presence/motion: AUROC, AUPRC, F1, false alarms per hour/day, detection delay.
- Zone: macro-F1 and median localization error.
- Count: MAE and confusion matrix for 0 / 1 / 2+.
- Welfare rule: event recall, false alerts per day, time-to-alert.
- Identity: TAR at fixed FAR, EER, open-set AUROC, F1 for “unknown,” and calibration error.

Vital signs

- MAE and RMSE in beats/min or breaths/min.
- Bland–Altman plots against reference.
- Percentage within predefined error bands.
- Error stratified by room, day, posture, distance, orientation, participant, device layout, LOS/NLOS, and signal-quality bin.
- Coverage-risk curve for abstention.
- Explicitly separate stationary and moving windows.

For respiration, a credible target is not a single headline number. VitalCSI reports $R^2 = 0.93$ and 1.20 breaths/min MAE across 6–33 breaths/min using nasal airflow reference in a controlled seated setup. That is a useful benchmark, but your unseen-room, free-living-style condition is harder and should not be expected to match it automatically. ^[18]

A practical 16-week roadmap

Weeks 1–2: literature and reproduction

- Read the 12 core papers.
- Prepare a comparison table: hardware, participants, rooms, task, ground truth, split protocol, metric, limitation.
- Reproduce one public benchmark baseline with SenseFi.
- Write your problem statement and preregistered evaluation protocol.

Weeks 3–4: CSI capture pilot

- Choose hardware:
 - Fastest low-cost path: 2–3 ESP32 nodes for presence/motion experiments.
 - Better vital-sign path: Raspberry Pi 4 + compatible Nexmon CSI capture and a modern 5 GHz router.
- Verify packet rate, packet loss, timestamp stability, raw-data format, and synchronization.
- Run 10–20 pilot sessions in one room.
- Do not begin deep learning before inspecting raw time series and spectra.

Weeks 5–7: first dataset

- Record controlled single-person sessions across rooms, days, positions, orientations, and layouts.
- Collect empty-room and non-human disturbance sessions: fan, curtain, door, pet if relevant, nearby person outside the room.
- Add synchronized respiration and heart-rate references.
- Create strict session-level splits before model training.

Weeks 8–10: baselines

- Implement motion/presence baseline.
- Implement respiration signal-processing baseline.
- Train CNN, GRU/LSTM, and small Transformer baselines.
- Establish leave-one-room-out and leave-one-person-out results.

Weeks 11–13: core contribution

- Add multi-source domain generalization.
- Add a quality/abstention head.
- Run ablations:
 - No DG.
 - DG only.
 - Quality only.
 - DG + quality.
 - With versus without self-supervised pretraining.
 - Raw amplitude versus CSI ratio / phase-difference representation.

Weeks 14–16: edge and paper

- Quantize the best model.
- Deploy it locally.
- Measure actual latency, memory, capture-to-decision delay, and energy.
- Run failure analysis rather than hiding bad conditions.
- Write dataset card, privacy statement, ethics/consent process, reproducibility checklist, and code/data release plan.

What not to claim

Avoid these statements unless you genuinely prove them with the right protocol:

- “Works in any room.”
- “Clinical vital-sign monitoring.”
- “Heartbeat provides a unique intruder biometric.”
- “Privacy preserving” without describing data retention, consent, access control, and the fact that RF sensing can infer sensitive behaviors.
- “Low power” without measured energy and latency.
- “Generalizable” without unseen-home, unseen-day, unseen-person, and unseen-layout tests.
- “Intruder detection” when you only evaluated closed-set classification among volunteers.

Your strongest final thesis contribution would be:

A reproducible, continuous, multi-home CSI dataset and quality-gated domain-generalization framework that reports when WiFi sensing works, when it fails, and when it should abstain—demonstrated for presence, movement, zone-level monitoring, respiration-aware stationary presence, and open-set gait-based authorization on constrained edge hardware.

That is narrower than the original plan, but much more credible, publishable, and useful.

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